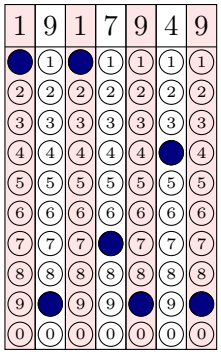
TEST VARIANTI RAQAMI: **1917949**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!



1. Agar $f(x) = 10 \cdot 6^x$ bo‘lsa, $f(-1) + f(-2) + f(-3) + f(-4) + ...$ ni hisoblang.

A) $\frac{215}{108}$ B) *2* C) $\frac{5}{3}$ D) *1*

2. To‘g‘ri burchakli parallelepipedning uchta turli yoqlarining diagonallari 7; 8 va 9 dm bo‘lsa, parallelepipedning diagonali uzunligini (dm) hisoblang.

A) $4\sqrt{6}$ B) $\sqrt{97}$ C) $\sqrt{94}$ D) $6\sqrt{3}$

3. Hisoblang:

$$\left(\frac{\sqrt{\sqrt{1009}+12\sqrt{7}}}{\sqrt{\sqrt{1009}-12\sqrt{7}}}+\frac{\sqrt{\sqrt{1009}-12\sqrt{7}}}{\sqrt{\sqrt{1009}+12\sqrt{7}}}\right)^2$$

A) *4036* B) *4038* C) *2019* D) *2018*

4. $15 \cdot 2^{21} \cdot 5^{17}$ ko‘paytma nechta nol bilan tugaydi?

A) *19* B) *18* C) *21* D) *22*

5. Quyidagi sonlardan nechtasi butun son?

1) $\sqrt{2+\sqrt{144}}$; 2) $7,12$; 3) $\pi - 1,14$; 4) $-\frac{2,48}{1,24}$

A) *2* B) *3* C) *4* D) *1*

6. $f(x) = \frac{x \cdot 2^x}{\ln 2} - \frac{2^x}{\ln^2 2} + 2x$ funksiyaning x_0=1 nuqtadagi hosilasini toping.

A) *2* B) *4* C) *−4* D) *−2*

7. Qirrasining uzunligi 6 cm ga teng bo‘lgan muntazam tetraedrning hajmini (cm^3) toping.

A) $32\sqrt{2}$ B) $18\sqrt{2}$ C) $24\sqrt{2}$ D) $36\sqrt{2}$

8. Agar $x^2 - \left(\sqrt{3}-2\right)x - \sqrt{4+2\sqrt{3}} = 0$ tenglamaning ildizlari x_1 va x_2 bo‘lsa, u holda $|x_1 - x_2|$ ni toping.

A) $\sqrt{3}$ B) $\sqrt{7}$ C) $\sqrt{2}$ D) $\sqrt{11}$

9. k ning qanday qiymatlarida $\frac{3x+2}{4x-3} = k+2$ tenglamaning ildizi 1 dan kichik bo‘ladi?

A) $(-3; 1,25)$ B) $(-\infty; -3) \cup (1,25; \infty)$
C) $(-\infty; -1,25) \cup (3; \infty)$ D) $(-1,25; 3)$

10. Amallarni bajaring: $\frac{1}{6x^3} - \frac{2}{3x^2y} - \frac{3}{4y^2}$

A) $\frac{2y^2-8x^2y-9x^3}{12x^3y^2}$ B) $\frac{2y^2-8xy^2-9x^3}{12x^3y^2}$
C) $\frac{2y^2-8xy-9x^3}{12x^3y^2}$ D) $\frac{2y^2-8xy-9x^2}{12x^3y^2}$

11. Agar $x = 17$ bo‘lsa, $\left(\frac{x\sqrt{x}-64}{x-16} + \frac{4\sqrt{x}}{\sqrt{x}+4}\right) : \left(\frac{8}{4-\sqrt{x}} - 1\right) - 4$ ifodaning qiymatini toping.

A) 4 B) $\sqrt{17}$ C) -4 D) $-\sqrt{17}$

12. $8 \cdot 2^{8x+5} = \sqrt[5]{16^{x+100}}$ tenglamani yeching.

A) *11* B) *9* C) *12* D) *10*

13. Hasan, Husan va ularning 3 nafar o‘rtoqlari orasidan ixtiyoriy tanlangan 3 kishining orasida Hasan va Husan bo‘lishi ehtimolligini toping.

A) *0,4* B) *0,25* C) *0,3* D) *0,2*

14. $|\vec{a}| = 3$ va $\left|\vec{b}\right| = \sqrt{2}$, $\vec{a}$ va $\vec{b}$ vektorlar orasidagi burchak $\frac{\pi}{4}$ ga teng. $2\vec{a} - \vec{b}$ va $3\vec{a} + 4\vec{b}$ vektorlarning skalyar ko‘paytmasini toping.

A) $46 + 15\sqrt{2}$ B) 42 C) 61 D) $46 + 5\sqrt{2}$

15. Nechta butun son $\sqrt[4]{2x^2+7} - \sqrt{x-2} < 0$ tengsizlikning yechimi bo‘ladi?

A) *0* B) *3* C) *2* D) *1*

16. Soatning minut mili 15 minutda necha gradusga buriladi?

A) *105°* B) *75°* C) *60°* D) *90°*

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17. $A = \{x \mid x \geq 2,\ x \in Z\}$ to‘plamning Z to‘plamgacha to‘ldiruvchi A' to‘plamini toping. $(A' = Z \setminus A)$

A) $A' = \{x \mid x \leq 1,\ x \in Z\}$

B) $A' = \{x \mid x \in Z\}$

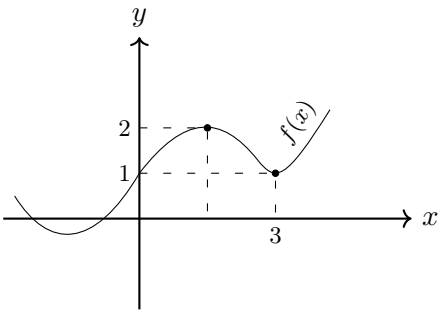
C) $A' = \{x \mid x \leq 2,\ x \in Z\}$

D) $A' = \{x \mid x < 1,\ x \in Z\}$

18. Ikki natural sonning EKUKi 168 ga teng va ularning nisbati 3:4 kabi bo‘lsa, kichik sonni toping.

A) *56* B) *48* C) *36* D) *42*

19. Rasmda $f(x)$ funksiyaning grafigi tasvirlangan. Agar $g(x) = (2x-7)^7 \cdot f(x)$ bo‘lsa, $g'(3)$ ni toping.



A) -14 B) -7 C) 7 D) 14

20. $ABCD$ parallelogrammning BC va CD tomonlaridan mos ravishda M va N nuqtalar shunday tanlab olinganki C uchidan boshlab hisoblaganda (BC va CD tomonlarini) 1:3 nisbatda bo‘ladi. Agar parallelogrammning yuzi 64 ga teng bo‘lsa, $AMND$ to‘rtburchakning yuzini toping.

A) *38* B) *40* C) *32* D) *36*

21. $25^{\log_5 x} - 5 \cdot 2^{\log_2 x} = 24$ tenglamaning ildizi x_0 bo‘lsa, $x_0^2 - 5x_0 + 7$ ning qiymatini toping.

A) *33* B) *31* C) *30* D) *32*

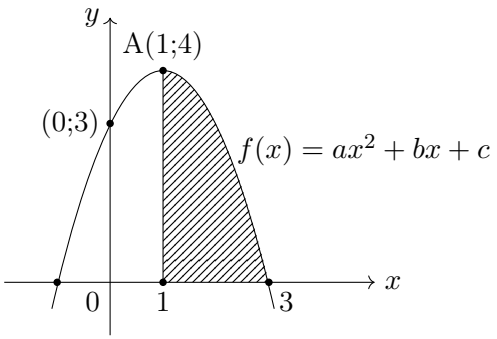
22. a, b, c sonlar uchun $7a = 2b = 3c$ va $\frac{3}{a} - \frac{5}{b} + \frac{2}{c} = 3\frac{2}{5}$ tengliklar o‘rinli bo‘lsa, $\frac{b}{c} - a$ ning qiymatini toping.

A) $-\frac{5}{7}$ B) $1\frac{1}{14}$ C) $-\frac{3}{14}$ D) $\frac{11}{14}$

23. Yon tomonining uzunligi 5 dm bo‘lgan teng yonli trapetsiyaga doira ichki chizilgan. Agar trapetsiyaning yuzi 20 dm^2 bo‘lsa, doiraning yuzini (cm^2) toping.

A) 16π B) 20π C) 400π D) 40π

24. Rasmda shtrixlangan soha yuzini toping. (A – nuqta parabolaning uchi)



A) $5\frac{1}{3}$ B) 9 C) $8\frac{1}{3}$ D) $6\frac{2}{3}$

25. $f(x) = x^2 - 2x + c$ funksiyaning nollaridan biri 3 bo‘lsa, ikkinchisini toping.

A) -1 B) 1 C) 0 D) -5

26. Hisoblang: $\sin 14^\circ + \cos 14^\circ \cdot \operatorname{tg} 38^\circ - 1$.

A) 1 B) 2 C) -1 D) 0

27. $y = -3x + 7$ chiziqli funksiyaning abssissalar o‘qiga nisbatan simmetrigini toping.

A) $y = 3x - 7$ B) $y = 3x + 7$
C) $y = -3x - 7$ D) $y = -3x + 7$

28. Agar $\sin\alpha = \frac{2}{5}$ bo‘lsa, $\cos^6\alpha - 3\cos^4\alpha + 3\cos^2\alpha + \sin^6\alpha - 1$ ning qiymatini toping.

A) 1 B) 0 C) -1 D) -2

29. Agar 8 litr dengiz suvida o‘rtacha 300 g tuz bo‘lsa, 5 m^3 dengiz suvida o‘rtacha necha kilogramm tuz bo‘ladi?

A) *224,5* B) *166* C) *187,5* D) *150,5*

30. Toza suvga tuz aralashtirilgandan so‘ng massasi 850 gramm bo‘ldi. Agar tuzning massasi toza suvning massasidan 75% ga kam bo‘lsa, toza suvning massasi necha gramm bo‘lgan?

A) *660* B) *640* C) *680* D) *620*

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Test varianti “Test topshiriqlari to‘plami 2019” asosida shakllantirilgan:

1 – 29-bet 1	7 – 109-bet 24	13 – 125-bet 23	19 – 81-bet 42	25 – 72-bet 27
2 – 111-bet 42	8 – 65-bet 61	14 – 119-bet 41	20 – 103-bet 97	26 – 37-bet 33
3 – 28-bet 61	9 – 63-bet 41	15 – 67-bet 8	21 – 58-bet 35	27 – 74-bet 46
4 – 24-bet 20	10 – 47-bet 27	16 – 91-bet 12	22 – 14-bet 50	28 – 41-bet 64
5 – 12-bet 27	11 – 48-bet 36	17 – 121-bet 13	23 – 96-bet 46	29 – 20-bet 37
6 – 77-bet 9	12 – 54-bet 26	18 – 9-bet 25	24 – 89-bet 37	30 – 19-bet 25

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